

Supplementary Material for FashionMAC: Deformation-Free Fashion Image Generation with Fine-Grained Model Appearance Customization

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In this supplementary material, we provide more details of the FashionMAC and additional results.

Garment-Centric Pose Predictor

Without any human structural priors, directly outpainting a deformed garment to synthesize a fashion model wearing it is very difficult. Therefore, we first design a garment-centric pose prediction model to generate the corresponding poses that fit the given garment.

Specifically, the garment-centric pose predictor is based on the LDM framework. Given an input fashion showcase image, we obtain the deformed garment image and the corresponding pose map from it by a pre-trained cloth segmentation model (Dabhi 2021) and a DensePose model (Güler, Neverova, and Kokkinos 2018), respectively. We then train a UNet denoiser ϵ_{θ_p} from scratch to generate the pose conditioned on the garment. Since the pose map and the garment image are spatially aligned, we simply concatenate the garment image’s latent encoding z_c with the pose image’s noised latent encoding z_{p_t} in the channel dimension as the input to the UNet denoiser ϵ_{θ_p} at timestep t . The objective function for training is as follows:

$$\mathcal{L}_{pose} = \mathbb{E}_{p,c,\epsilon \sim \mathcal{N}(0,1),t} [\|\epsilon - \epsilon_{\theta_p}(z_{p_t}, z_c, t)\|_2^2]. \quad (1)$$

During the inference, thanks to the stochasticity of the reverse denoising process, we can utilize the garment-centric pose predictor to sample diverse pose maps that fit the given garment.

Dataset

To further evaluate the generalization ability of our method, we conduct experiments on the IGPair (Shen et al. 2025) dataset. The original dataset contains over 300,000 training pairs and 2,000 testing pairs. To adapt it to our task, we construct a curated subset. Specifically, we select six garment categories based on the labels, including ‘upper_body’, ‘dresses’, ‘skirts’, etc. We then perform de-duplication to ensure that each garment image corresponds to a single fashion

showcase image. To guarantee the quality of facial generation, we filter the dataset to retain images with a sufficient face resolution (specifically, a face area ratio between 0.011 and 0.05). Furthermore, we adjust the pose distribution by decreasing the proportion of images with side or back views. Consequently, we obtain a dataset comprising 24,628 training pairs and 2,000 testing pairs for our experiments. The models are trained at a resolution of 512×640 .

To support our framework, we construct fine-grained text prompts and corresponding spatial masks for each training image of our two datasets. For the prompt extraction, we adopt QWen 2.5VL (Yang et al. 2025) to generate descriptive attribute prompts for predefined semantic regions. We define 9 core region categories that are most relevant to fashion, including expression, skin, hair, mouth, etc. To extract the spatial region masks, we utilize the semantic segmentation maps from our datasets for the body region and employ additional segmentation tools SAM (Kirillov et al. 2023) and Bisenet (Yu et al. 2018) to obtain the hair and face region.

Implementation Details

We adopt a three-stage training strategy: 1) We train FashionMAC without RADA for 55,000 iterations, using a batch size of 16 and learning rate of $2e-5$. 2) We add the RADA module and continue finetuning with ground-truth region masks for 6,000 iterations, using the same learning rate. 3) We freeze the denoising model and RADA, and train the mask prediction head for 60 epochs with a learning rate of $1e-4$. In inference, we replace ground-truth masks with predicted ones and apply the chained mask injection strategy to propagate structural guidance across timesteps.

Results Conditioned on Virtual Faces

Our FashionMAC can generate fashion showcase images under the guidance of facial images. To avoid issues of portrait rights of real humans, the input facial images can be obtained from portrait generation approaches such as StyleGAN (Karras, Laine, and Aila 2019). Fig. 1 demonstrates the results of FashionMAC conditioned on virtual faces generated by StyleGAN. The generated images successfully preserve the distinctive characteristics of the input faces, showcasing FashionMAC’s ability to maintain high

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fidelity in face-conditioned image generation.



Figure 1: The results of our method conditioned on virtual facial images generated by StyleGAN.

Qualitative Results of the Ablation Study

More qualitative results of the ablation study are shown in Fig. 2 to assess the effectiveness of the proposed Chained Mask Injection (CMI) strategy and Region-Adaptive Decoupled Attention (RADA) module. The ablation studies are conducted on our full model FashionMAC, model without CMI and model without CMI and RADA. These results further validate the effectiveness of CMI and RADA.

Experiments of the Mask Prediction

To enable accurate region-aware conditioning during the denoising process, we explore where to place the mask prediction head for optimal spatial guidance. We experimentally found that applying mask prediction on low-level encoder features at early denoising timesteps tends to produce imprecise region masks. In contrast, decoder-based masks remain relatively stable and less affected by the timestep. The results are illustrated in Fig. 3. We attribute this to the noisy nature of early-step features and the limited capacity of shallow encoder blocks to extract meaningful spatial cues from them. Conversely, decoder features benefit from deeper hierarchical feature integration, which enhances their ability to localize fine-grained semantic regions. Based on this observation, we adopt a chained mask injection strategy that progressively predicts and injects masks at network layers across timesteps, providing stable and precise regional priors to guide generation. The mask generation introduces only a marginal computational overhead—approximately 32.7 MFlops, corresponding to a 7% increase in inference time.

Robustness Analysis of Segmentation

The primary goal of FashionMAC is to generate high-quality garment showcase images for e-commerce retailers, rather than performing traditional Virtual Try-On (VTON). The critical objectives of this task are (i) preserving garment details and (ii) enabling controllable model appearance customization. Unlike standard VTON methods, our approach does not require generating garments from scratch; instead, it utilizes garment images directly segmented from the source models. Thanks to the powerful capabilities of modern segmentation models (e.g., SAM 3 (Carion et al. 2025)), our method receives reliable semantic conditioning in most cases. Furthermore, for occasional imperfect segmentations, users can manually improve the semantic maps using editing tools (e.g., Photoshop) to mitigate potential artifacts.

To further evaluate the impact of segmentation quality on our results, we conducted experiments by degrading the input garment images to simulate varying degrees of segmentation artifacts. The results are illustrated in Fig. 4. We observe that even under extreme cases, our method is capable of maintaining the overall appearance to some extent. However, faithfully preserving specific details within the missing regions remains challenging. As the segmentation quality increases, the alignments between the generated images and the target garment are improved.

More Results of the Fine-grained Customization

Fig. 5 and Fig. 6 present additional qualitative comparisons and diverse visual results of FashionMAC. These examples highlight the framework’s superior capability for fine-grained appearance customization and its effectiveness in generating high-fidelity, diverse fashion showcase images.

Generation Conditioned on Hard-set Masks.

Our model can generate fashion images conditioned on hard-set spatial masks. The results are presented in Fig. 7, which demonstrates the ability of FashionMAC to faithfully adhere to the specified region during generation. It further validates the effectiveness of our proposed Region-Adaptive Decoupled Attention (RADA) strategy, confirming that FashionMAC is capable of supporting fine-grained, customized image generation.

Experiments on the IGPair Dataset

Baselines. We compare FashionMAC with MagicClothing (Chen et al. 2024) and IMAGDressing (Shen et al. 2025) on the selected IGPair subset. To ensure a fair comparison, we retrain the baseline methods on the same IGPair subset.

Comparison. Fig. 8 presents the qualitative comparisons between FashionMAC and the baseline methods. The left panel displays the results generated with only text prompts, while the right panel shows the results under facial image guidance. For the results conditioned solely on text prompts, our method achieves superior text-image consistency compared to the baselines, while simultaneously maintaining

Methods	With facial image guidance				With text prompts guidance		
	FID↓	KID↓	CLIP-i↑	MP-LPIPS↓	FID↓	KID↓	MP-LPIPS↓
MagicClothing	39.17	0.0213	0.81	0.178	23.36	0.0086	0.167
IMAGDressing	20.27	0.0040	0.89	0.164	16.97	0.0047	0.155
Ours	10.23	0.0026	0.96	0.146	13.10	0.0040	0.146

Table 1: Quantitative comparison on the IGPair dataset.

high garment fidelity. Regarding the results under facial image guidance, our method demonstrates the ability to faithfully preserve the given facial identity. Tab. 1 shows the quantitative comparison results between FashionMAC and the baseline methods. Our method outperforms MagicClothing and IMAGDressing across all evaluation metrics. These results further demonstrate the superior capability of FashionMAC.

Comparison with the Virtual Try-On Based Methods

To evaluate the performance of our method and virtual try-on (VTON) based methods on the task of fashion image generation, we conduct an experiment with a two-stage VTON-based pipeline CosmicMan+TPD. In the first stage, we generate a virtual fashion model to show the target garment using a state-of-the-art text-to-image human generation framework CosmicMan (Li et al. 2024). In the second stage, we utilize the VTON method TPD (Yang et al. 2024) to generate the showcase image in which the above virtual fashion model wears the target garment. This pipeline mimics the showcase image generation capabilities of our framework. The results reveal that FashionMAC achieves outstanding generation performance, excelling in preserving human structural features and garment fidelity. Fig. 9 demonstrates some generated showcase images with CosmicMan+TPD and our method, respectively.

Fashion Image Generation for Mannequins

In e-commerce, the retailers usually use mannequins to coarsely demonstrate the garments. Our FashionMAC can utilize mannequin garment images as input and generate realistic apparel showcase images. For an input mannequin image that wears a garment, we can obtain the target garment through a pretrained segmentation network. Then FashionMAC can generate the corresponding pose map and produce high-quality fashion images. Fig. 10 shows the results conditioned on mannequin images. The results indicate the robustness and generalization ability of our method.

Applicability of FashionMAC

The goal of FashionMAC is to generate high-quality garment showcase images for e-commerce retailers. The key points of the task are (i) preserving garment details and (ii) controllable model appearance customization. Given a fixed segmented garment, the diversity of body shapes and the poses of the synthesized models are limited. Yet, FashionMAC can accept photos of users or plastic mannequins

wearing the garment with diverse body shapes and poses as inputs, and then customize the model appearances from these inputs, which are easy to obtain in real e-commerce scenarios. Currently, FashionMAC’s performance across different body shapes and poses is constrained by limited dataset diversity. However, as FashionMAC does not require paired training data as in VTON, expanding training data to include a wider range of body shapes should substantially improve its generalization.

Limitation and Future Work.

Since current fashion datasets contain data biases, such as the predominance of young female fashion models, the range of controllable attributes achievable by our method trained on these datasets is limited. Expanding these datasets to include a broader variety of ages, body types, genders, cloth types and backgrounds would be a promising direction for future work, as it would enhance the versatility and controllability of our method in generating more diverse and representative outputs.

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Figure 2: The qualitative results of the ablation study.

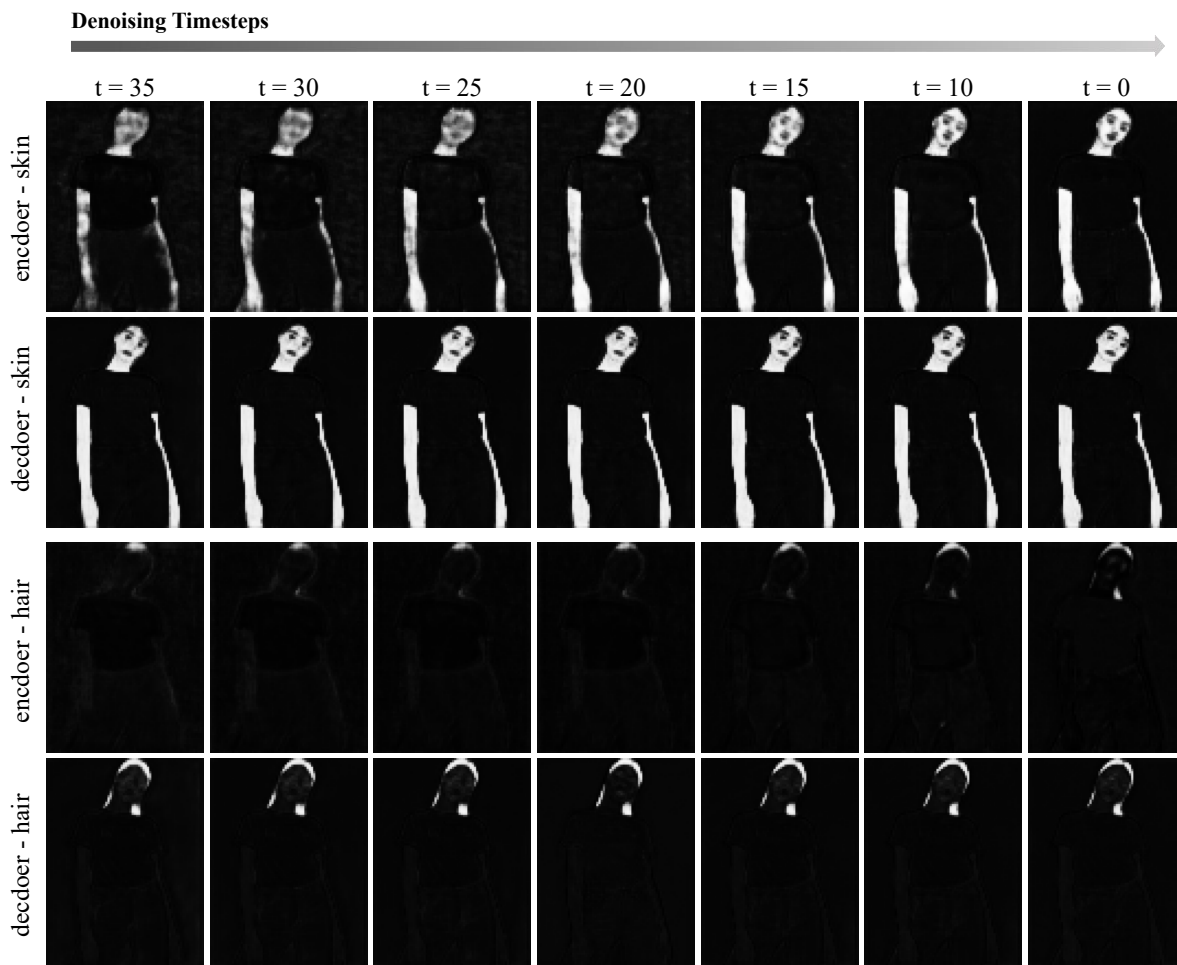


Figure 3: The masks predicted by different parts of the network at different timesteps.



Figure 4: Results for the same garment under varying segmentation quality.



Figure 5: Qualitative comparison with existing methods.



Figure 6: More Results of FashionMAC.



Figure 7: The results conditioned under hard-set masks for the hair region.



Figure 8: Comparison with the baseline methods on the IGPair subset. The left side shows the results with text prompt guidance. The right side shows the results with facial image guidance.

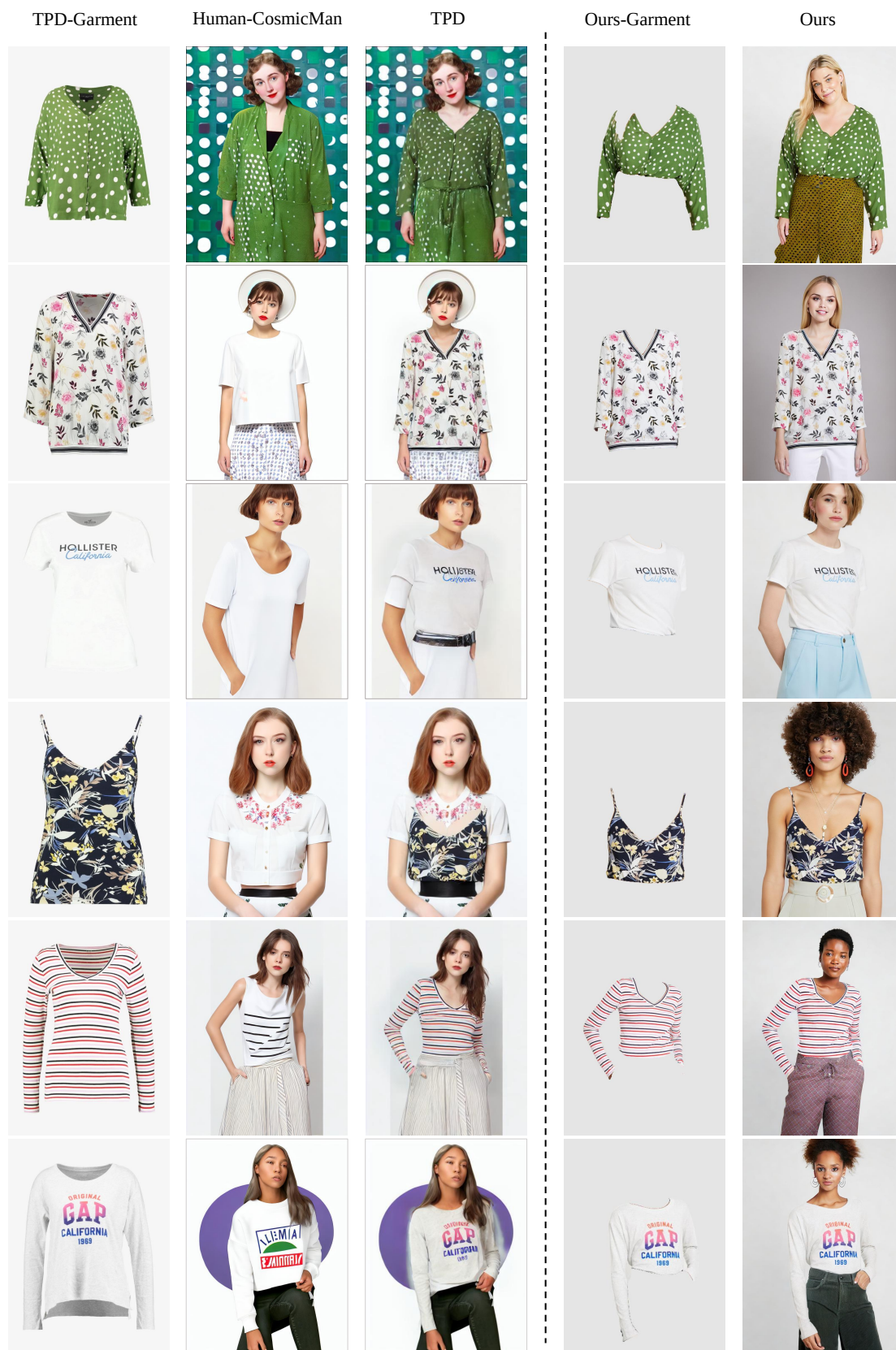


Figure 9: Qualitative comparison with CosmicMan+TPD.



Figure 10: The results of our method conditioned on garments worn by a mannequin.